

Continental shelves

The oceans are fringed by coastal seas that are much shallower than the open oceans. This is because the seabed here is not the deep ocean floor, but the flooded edge of a continent. The edge is cut away at sea level by wave erosion, creating a shallow seabed of continental rock. This is the continental shelf. At its outer edge, it falls away as the continental slope, which descends to the ocean floor.



COASTAL EROSION

The edges of continents are eaten away by the sea waves in a relentless process that turns solid rock into the shingle and sand that form beaches. This coastal erosion creates the shallow seabed of the continental shelf. It is made of the same solid bedrock as the land, covered with sedimentary rock.

WOW!

In the Scandinavian Arctic, the continental shelf extends most of the way to the North Pole.

SHELF, SLOPE, AND RISE

On average, the continental shelf extends about 50 miles (80 km) from the coast. The outside edge is called the shelf break, and beyond this, the gradient falls away as the continental slope. At the foot of the slope, a layer of rocky debris called the continental rise hides the transition from continental rock to the basalt of the true ocean floor.

